

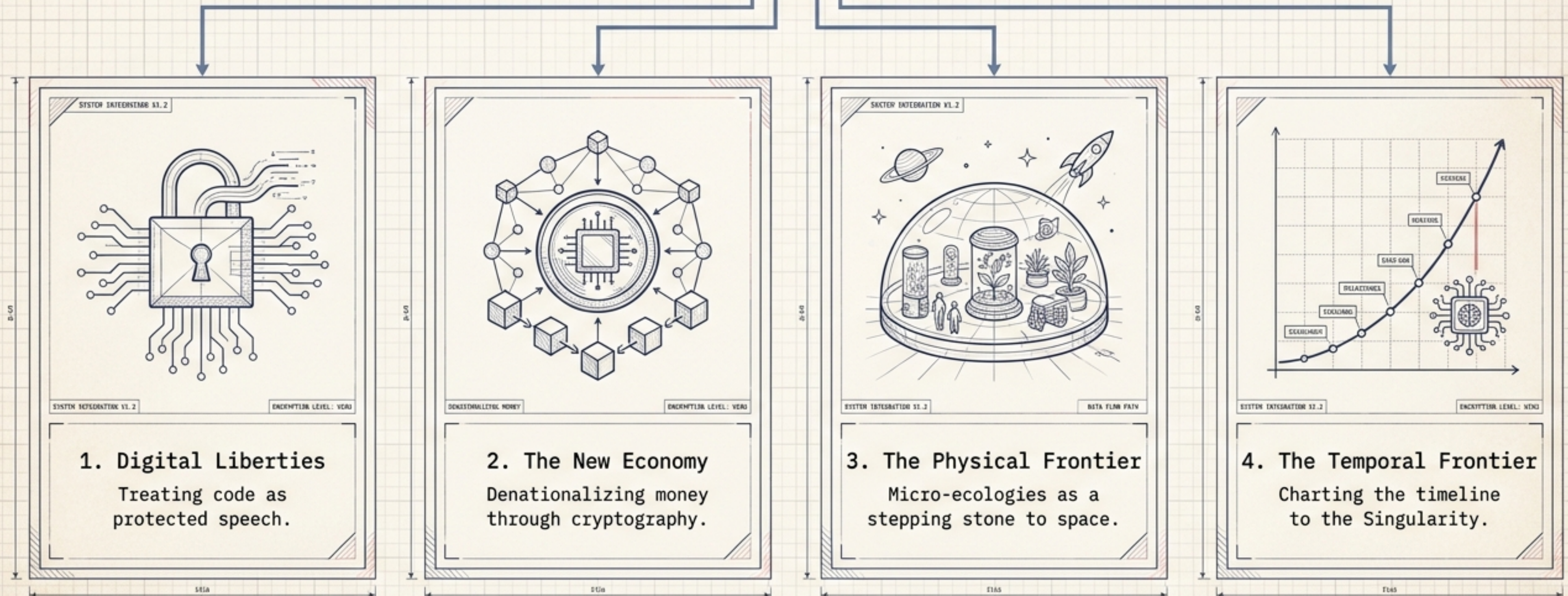
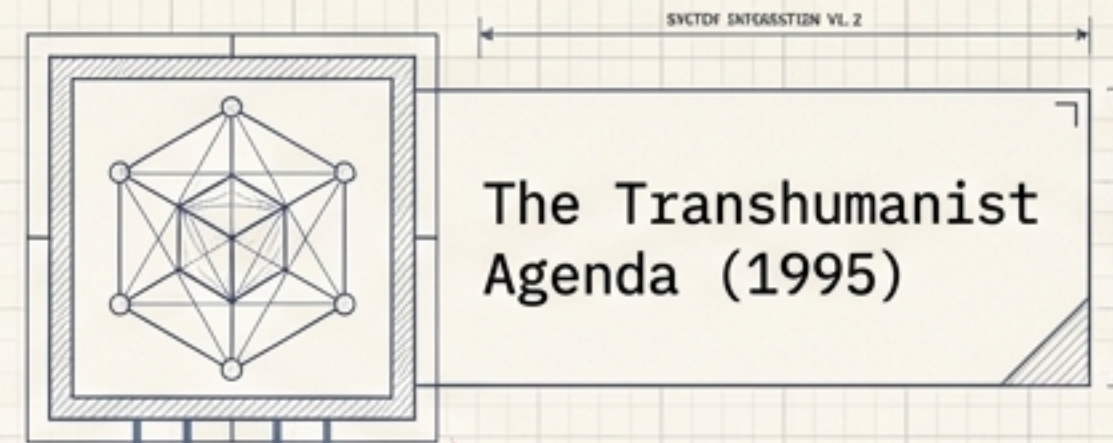
The Cyber-Blueprint: Visions from the Transhuman Vanguard

A retrospective synthesis of Extropy Magazine (1995) and the foundational ideas that mapped the frontiers of digital liberty, decentralized economics, and artificial ecology.



Mapping the Frontiers of 1995

Long before the mainstream adoption of cryptocurrency or the commercialization of space, the thinkers behind Extropy were reverse-engineering the limits of human capability. Their blueprint for the future rested on four hackable systems:



Code is Human-to-Human Communication

The Threat

- ⚙️ **Software Patents:** "Administrative fictions" born from a misreading of the 1981 Diamond v. Diehr case.
- 🖨️ **Export Controls:** The U.S. government regulating cryptographic software as "a tool of war."
- 🔒 **Censorship:** Apache developers forced under threat of felony prosecution to remove code "hooks" for encryption from the web.

The Workaround

- **Phil Salin's 1991 Doctrine:** Programs should be protected against any attempt by government to license what can be written.
- **The Print Loophole:** PGP and EFF's Cracking DES were published in physical, paper books to bypass digital export bans, allowing overseas users to legally scan and republish the code.

The Denationalization of Money

Based on the work of 1974 Nobel Laureate Friedrich Hayek, the Extropian vanguard recognized that a state monopoly on currency causes inflation, economic instability, and arbitrary geographic boundaries.

The Problem: Governments cover deficits by expanding the money supply, acting as a hidden tax that distorts relative prices and triggers boom/bust cycles.

State
Monopoly

The Solution:
Competing private currencies.

Polycentric
Private
Currencies

Polycentric
Private
Currencies

Polycentric
Private
Currencies

Polycentric
Private
Currencies

Polycentric
Private
Currencies

Polycentric
Private
Currencies

The Catalyst: Anonymous digital cash. Stripping the state of its ability to track and tax every transaction renders economic nationalism obsolete, paving the way for supranational free markets.

The Architecture of Exchange

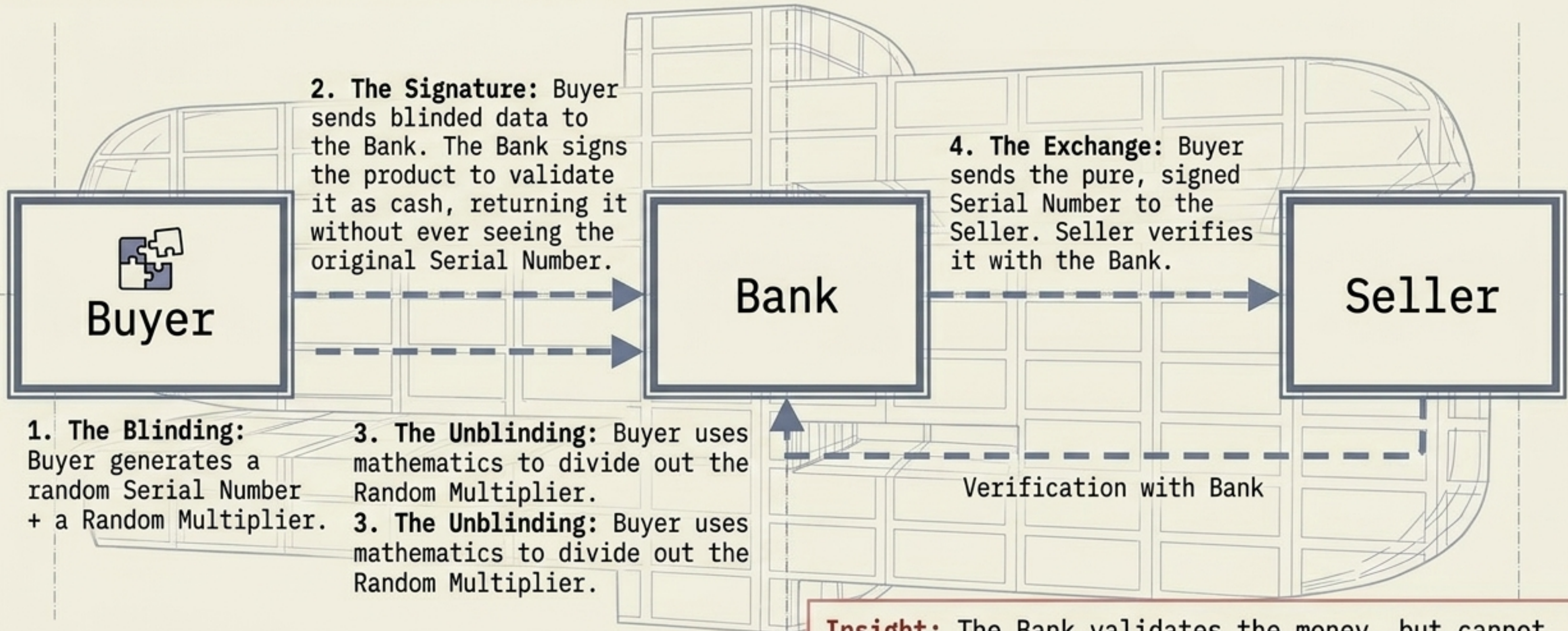
Privacy vs. Utility Matrix

	Anonymity	Double-Spend Risk	Friction (Time/Space)	Central Tracking
Physical Cash	High	Low (hard to forge)	High (requires physical proximity)	None
Credit / Checks	Zero (fully profiled)	Zero (bank verified)	Low	Absolute (creates marketing/gov trail)
True Digital Cash	High	Solved via Cryptography	Zero (Internet speed)	Blocked

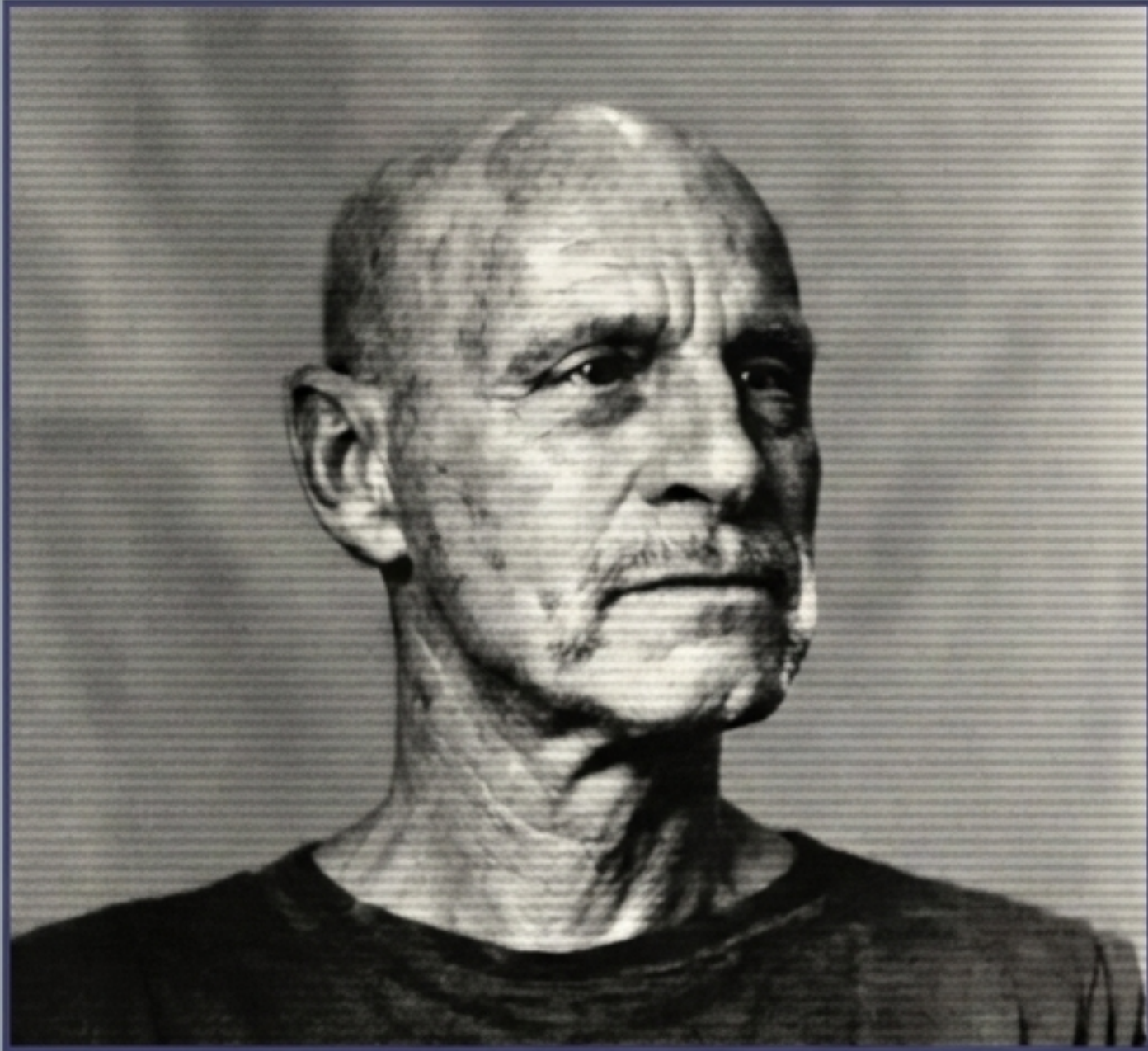
Many payment systems proposed for the Information Superhighway have more in common with checks than cash. What we really need is true digital cash. — Mark Grant

How to Prevent the Double-Spend

Because digital cash is just data, it can easily be copied. To prevent double-spending without destroying anonymity, David Chaum invented Blind Signatures.



Insight: The Bank validates the money, but cannot link the Serial Number back to the Buyer's withdrawal. The identity is mathematically sealed.



Dr. Roy Walford

Mission Dossier

Subject: Roy Walford, M.D. (UCLA Gerontology Expert).

Mission: Two continuous years sealed inside Biosphere 2 in the Arizona desert.

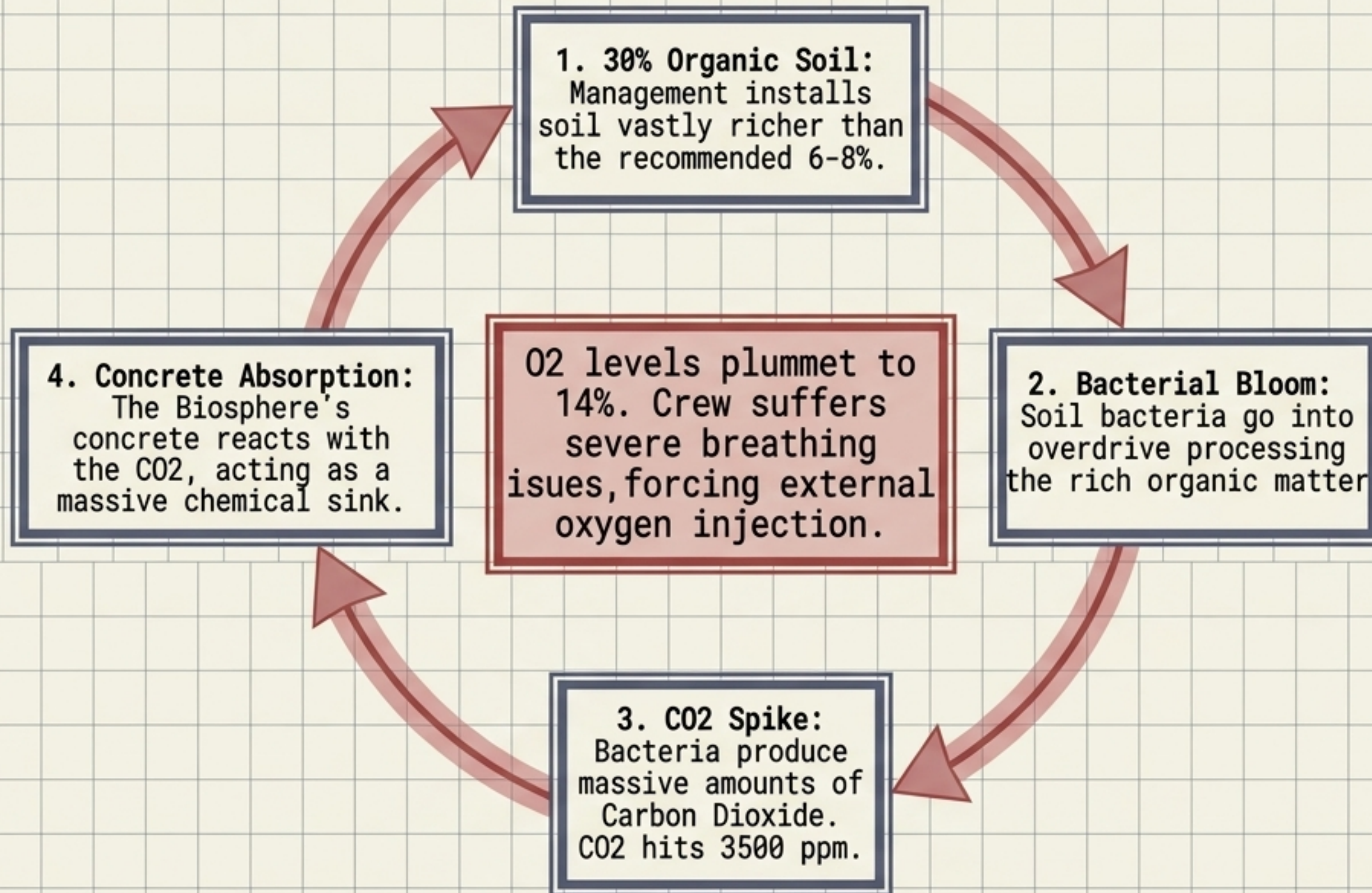
Goal: To study closed micro-ecologies as the necessary precursor to Martian colonization and O'Neill-style space habitats.

The Philosophy: "It's a serious mistake to wait till all the problems on Earth are solved before we go into space, because in that case we'll never get there."

Result: Though technically successful in remaining sealed without outside food, the human and ecological chaos inside proved that biology is profoundly harder to program than code.

The Oxygen Collapse: An Ecological Cascade

Against scientific advice, management insisted on using hyper-rich soil to ensure plant growth. This single variable triggered a catastrophic chemical loop.



Chaos Dynamics in Closed Environments

Variables in a Closed System

Biological Chaos

The Morning Glory Effect: Two thumb-sized vines were planted on a cliff wall. Liking the environment too much, they exploded, overgrowing the entire rainforest and space frame, requiring endless hours of manual hacking to control. (Australian cockroaches also thrived unexpectedly).

Sociological Chaos

- The Human Element: The crew fractured into two factions: four loyal to the outside management, and four objective scientists.
- Psychological Output: Routine psychometric testing (MMPI) failed because the crew's relentless determination to achieve the mission overrode standard psychological breakdown models.

Small, isolated space colonies will require exactly precise selection—one wrong biological or human variable radically alters the entire system's trajectory.

Forecasting the Singularity

To understand the trajectory of these technological vectors, Extropy assembled a vanguard of futurists (including Eric Drexler, Gregory Benford, and Nick Szabo) to predict the timeline of human transcendence.

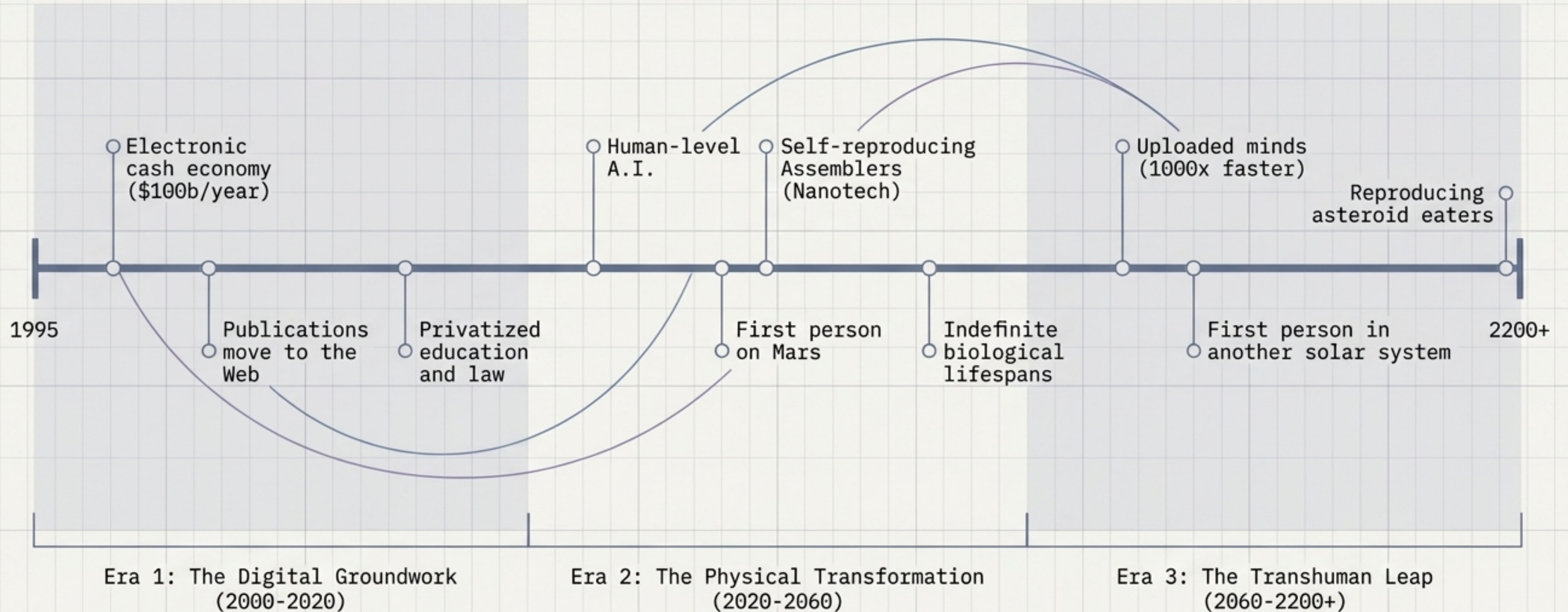
The Variables of Prediction

The Singularity (S): The point at which molecular assembly capability is achieved. Placed generally between 2005 and 2035.

The Design Ahead Factor (DAF): Mark Miller's formula accounting for technological acceleration. The sooner the Singularity occurs, the less "design ahead" will have anticipated it, causing a wider spread of subsequent breakthroughs.

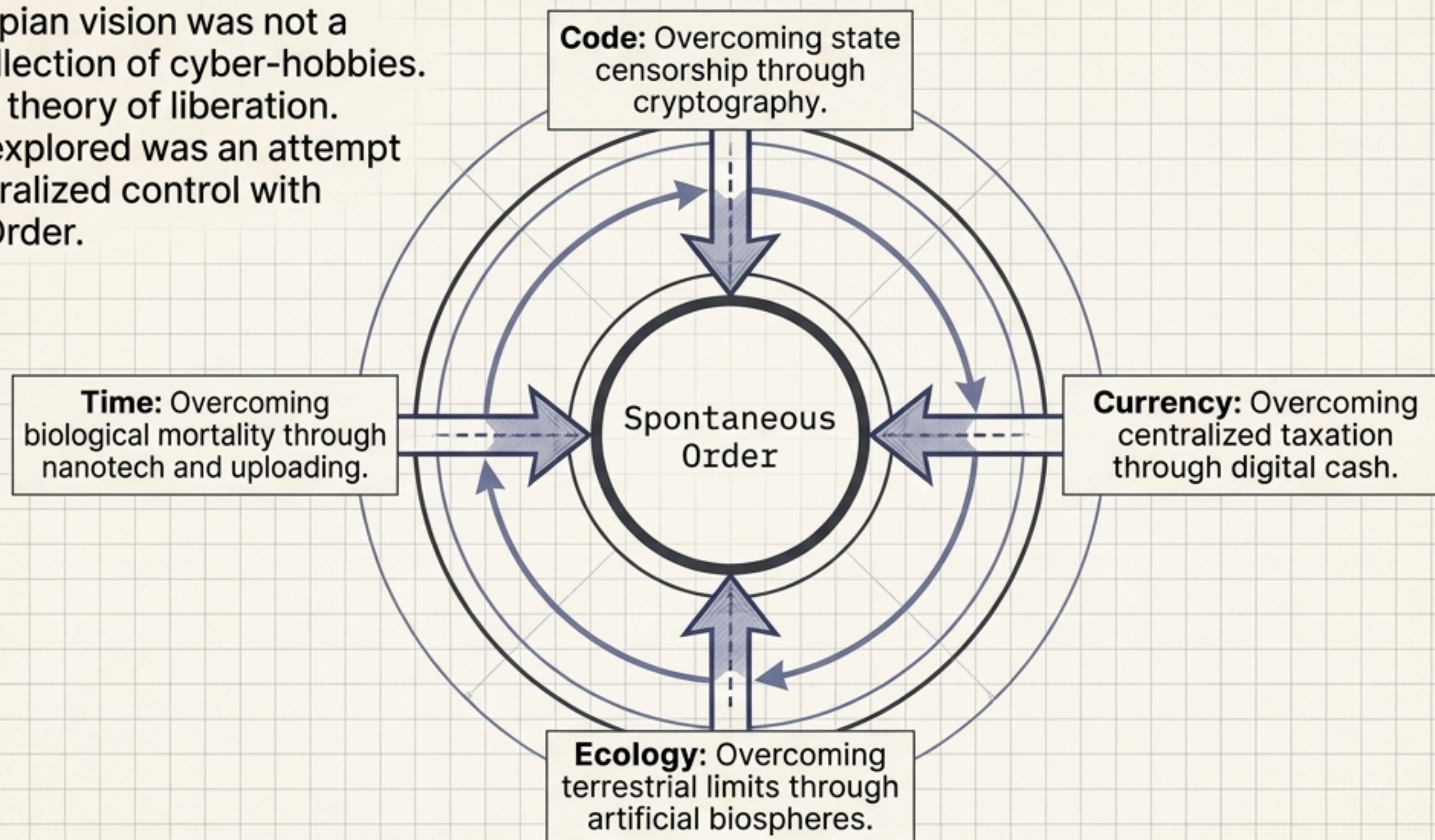
Future Forecasts	Molecular Date	Molecular Age	Molecular Age	Singularity Date	Predicts Thcoax! Psadiction	Nlugh	
Future Forecasts	Drexler	128	3/10/76	10 2035/97	2005	0.6243	13,406
Future Forecasts	Drexler	127	3/10/96	39 2020/50	1100	--	21,807
Future Forecasts	Bendess	289	13/30/76	20 2033/75	2400	0.4461	4,309
Future Forecasts	Benford	523	12/10/95	19 2082/35	21300	0.2733	24,341
Future Forecasts	Szabo	278	12/10/96	38 2003/85	71360	0.5034	24,870
Future Forecasts				15 2005/35	51360	0.2954	14,406
Future Forecasts				19 2005/85	71300	0.3945	16,592
Future Forecasts				25 2006/85	81300	0.2963	14,382
Future Forecasts				19 2005/35	81800	0.3943	14,895
Future Forecasts				18 2008/75	91800	0.1983	14,306
Future Forecasts				16 2058/93	101800	0.2968	14,960
Future Forecasts				36 2033/85	111360	0.1084	24,895
Future Forecasts				16 2008/75	121290	0.3943	28,505
Future Forecasts				16 2032/70	231300	0.2964	28,227
Future Forecasts				17 2035/85	121360	0.2954	23,905
Future Forecasts				18 2033/86	181560	0.2984	27,503
Future Forecasts				26 2033/03	281760	0.3983	20,402
Future Forecasts				19 2033/92	191700	0.2983	20,733
Future Forecasts				35 2033/85	141360	0.2084	26,808
Future Forecasts				11 2031/85	181400	0.2963	20,709
Future Forecasts				13 2033/92	251260	0.1984	26,506
Future Forecasts	Drexler	538	22/10/96	13 2030/88	241360	0.1561	20,050
Future Forecasts	Benford	258	23/18/79	13 2035/95	111260	0.1584	26,439
Future Forecasts	Drexler	376	23/18/36	13 2006/13	61509	0.1581	30,766
Future Forecasts	Benford	278	23/18/96	34 2009/85	31800	0.3764	20,905
Future Forecasts	Drexler	380	13/19/95	15 2009/03	71500	0.2783	24,796
Future Forecasts	Benford	276	23/16/35	16 2009/93	31800	0.3556	10,796
Future Forecasts	Szabo	288	53/15/96	35 2009/90	71800	0.2381	33,900
Future Forecasts	Szabo	380	16/15/93	36 2085/16	800	--	33,567
Future Forecasts	Szabo	343	13/15/96	13 2039/99	71560	0.2783	21,212
Future Forecasts	Benford	307	22/10/95	19 2039/33	301800	0.3701	33,359
Future Forecasts	Benford	397	24/15/96	30 2039/15	191500	0.2753	31,255
Future Forecasts	Szabo	378	16/18/96	8 2038/13	261560	0.1584	26,506

The Consensus Trajectory (1995 Forecast)



The Transhumanist Blueprint

The 1995 Extropian vision was not a fragmented collection of cyber-hobbies. It was a unified theory of liberation. Every frontier explored was an attempt to replace centralized control with Spontaneous Order.



The future is not a place we go; it is a system we engineer. Boundless expansion requires breaking the source code of society, economics, and biology itself.